Technische Hochschule Augsburg

STATISTICS

International Information Systems, 3rd semester

(3) Introduction to R

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R PROGRAMMING LANGUAGE

- Language and environment for statistical computing and graphics
  - Available as Free Software (GPL License)
  - Compiles and runs on a wide variety of UNIX platforms, Windows and MacOS

- Highly **extensible** to cover a wide variety of statistical and graphical techniques
  - can be expanded by downloading packages with **specific functionalities**
  - packages can be downloaded from CRAN (Comprehensive R Archive Network) and installed

- Has local user communities worldwide where you can network, share ideas, and learn
  - as new statistical techniques are developed, contributors react quickly to produce new R packages

https://cran.r-project.org/

R PACKAGES WE WILL USE

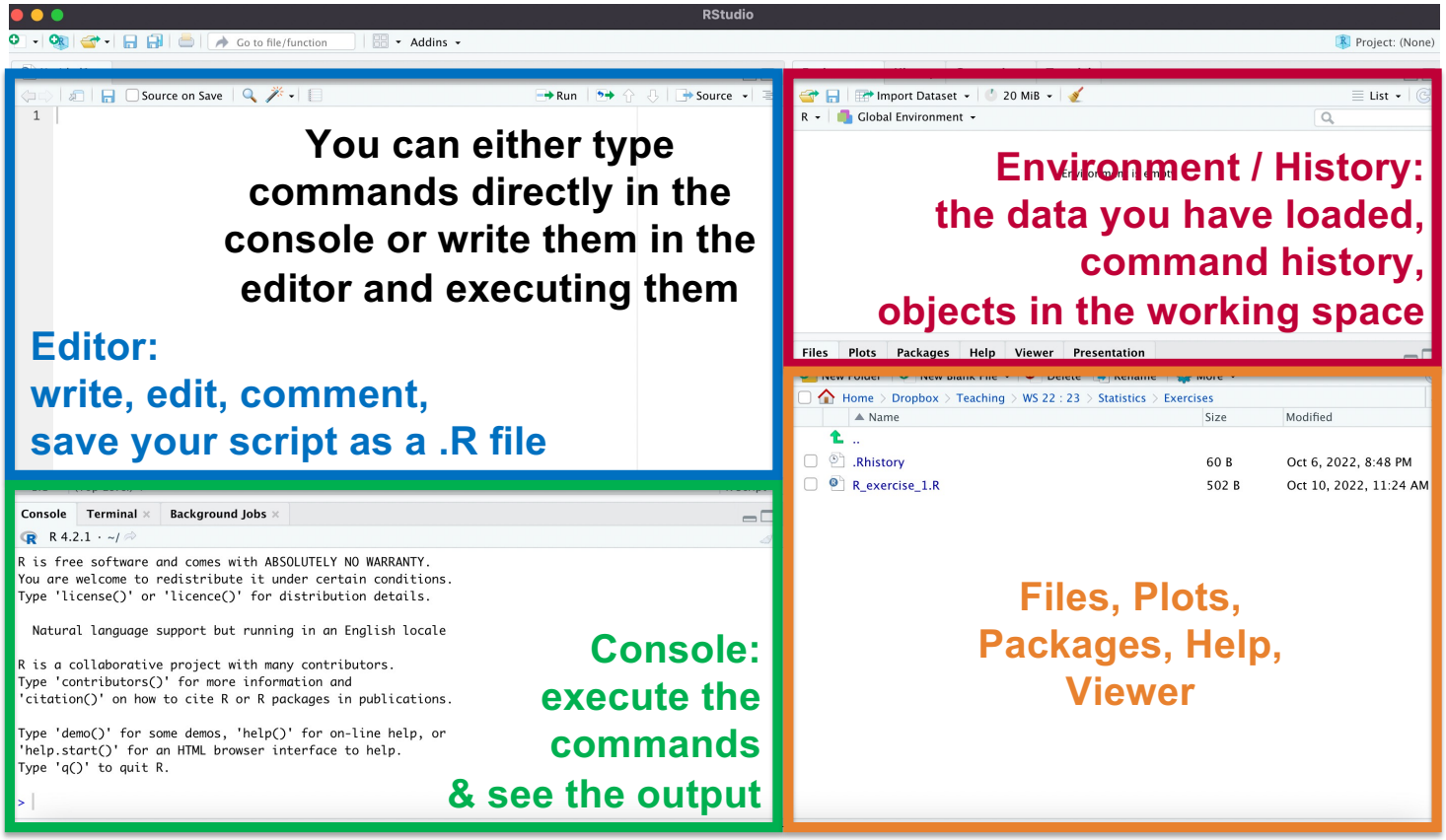
- **swirl:** a package to “learn R in R”
- **ggplot2:** a graphics package to create plots and graphs just with a few lines of code
- the **tidyverse:** packages for data manipulation, exploration and visualization sharing a common design philosophy
  - **tidyverse, dplyr, moderndive**
- some packages we will use mostly because of the datasets they come with
  - **languageR, ISLR, ISLR2, palmerpenguins**

R STUDIO

- In R you need to work with a **command line**
- R Studio provides a **useful interface** to use R
- It is an integrated development environment (IDE) for R
  - with command completion, syntax highlighting, documentation
  - available under the AGPL v3 open source license
- Two formats:
  - RStudio Desktop (which we will use) is a regular desktop application
  - RStudio Server (which we will not use) runs on a remote server and is accessed using a web browser

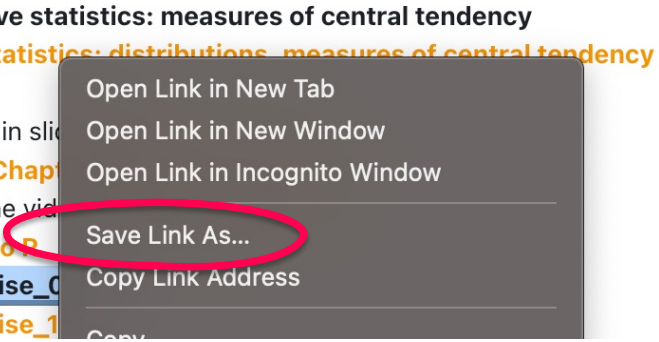
https://posit.co/download/rstudio-desktop/

R STUDIO


RStudio
You can either type commands directly in the console or write them in the editor and executing them
Editor:
write, edit, comment,
save your script as a .R file
Environment / History:
the data you have loaded,
command history,
objects in the working space
Console: execute the commands
& see the output
Files, Plots, Packages, Help, Viewer

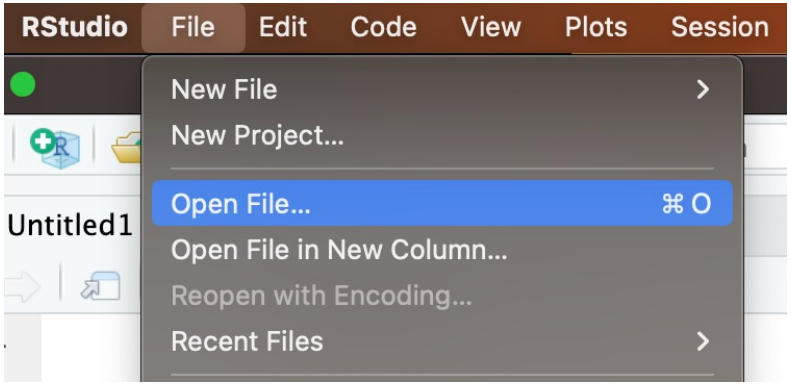
LET'S GET STARTED WITH R

Step 1: Save the .R files from Moodle



R_exercise_0.R

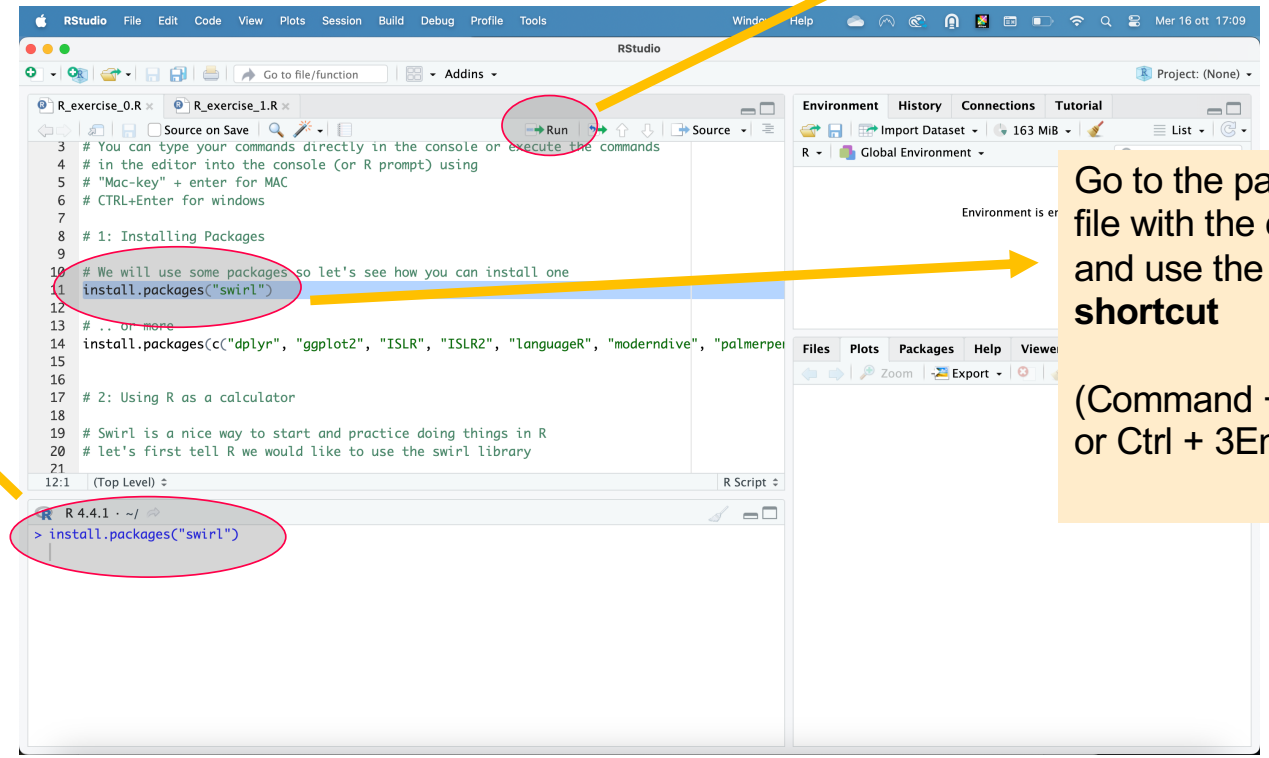
Step 2: Open them with RStudio



LET'S GET STARTED WITH R

Step 2: Open them with RStudio

With **run** you execute the file all together



Type the command in the **console**

Go to the part of the file with the command and use the **keyboard shortcut**

(Command + Return or Ctrl + 3Enter)

LET'S GET STARTED WITH R

Step 3: Downloading and installing packages

- Follow the comments and execute the commands in the file for "1: Installing Packages"
- To execute a command, go to the part of the file with the command (the first one is `install.packages("swirl")`) and use the keyboard shortcut
  - Command + Enter for MAC
  - CTRL + Enter for windows
- This command will install all the packages you will need:

```
install.packages(c("dplyr", "ggplot2", "ISLR", "ISLR2", "languageR",
"moderndive", "palmerpenguins", "swirl", "tidyverse"))
```

LET'S GET STARTED WITH R

Step 3: Downloading and installing packages

- You can use **packages, e.g. swirl, palmerpenguins**
- **install.packages("swirl")** → to install a package
- **installed.packages()** → to return a list of packages installed
- **find.package("swirl")** → to check if the package is installed and find its location
- **library(swirl)** → to load a package
  - You will only have to ***install*** the package once, but you will have to ***load*** it <u>every time you need it</u>

LET'S GET STARTED WITH R

{swirl}

Step 4: Learn the first steps with swirl

10-15 minutes

- Load the package using `library(swirl)`
- Follow the instructions that swirl gives you **as a "chatbot"** in the console

```
| Welcome to swirl! Please sign in. If you've been here before, use the same name as you did
| then. If you are new, call yourself something unique.

What shall I call you?
```

- Pick the course "R Programming" and the lesson "Basic Building Blocks"
- Try to understand the exercises it asks you to do
- Follow the comments and execute the rest of the commands in the file

LET'S GET STARTED WITH R

Step 5: Now let's look at a dataset in R

- The penguin dataset has been made available by Allison Horst as part of the **`palmerpenguins`** library https://allisonhorst.github.io/palmerpenguins/
- It contains data collected and made available by Dr. Kristen Gorman and the Palmer Station, Antarctica LTER, a member of the Long Term Ecological Research Network
- Two versions of the dataset are available
  - **`penguins`** (a simplified version of the raw data)
  - **`penguins_raw`** (with all the variables and original names as downloaded)

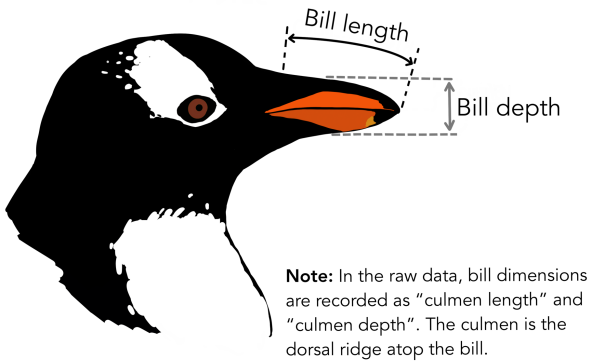
Artwork by @allison_horst

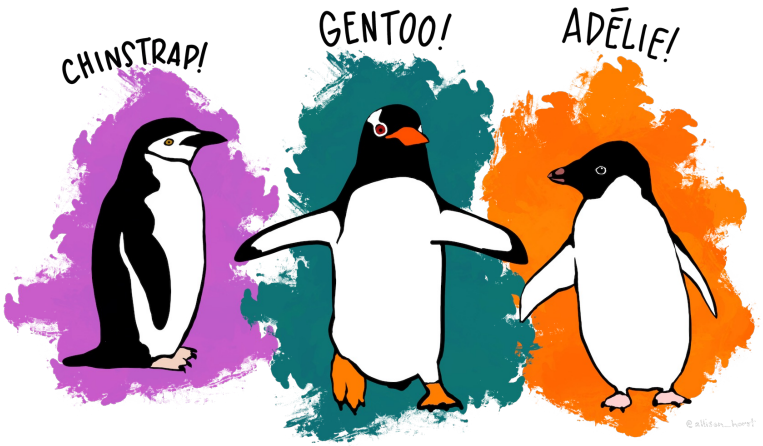
LET'S GET STARTED WITH R

Step 5: Now let's look at a dataset in R

- Both datasets contain data for 344 penguins
  belonging to 3 different species
  collected from 3 islands in the Palmer Archipelago, Antarctica
- Download and open the R_exercise_1.R file
- Load the **`palmerpenguins`** library
- Follow the comments and execute
  the commands in the file

R_exercise_1.R





Artwork by @allison_horst

SUMMING UP

- You can use R as a **calculator**
- You can create **variables** and store the results of operations and the output of functions in them
- You can create **vectors** as concatenations of values and perform operations with them
- Almost everything in R is done through **functions**
- **Data frames** are basically R's version of a spreadsheet.
  A data frame is a two-dimensional object, with rows and columns. Each column contains a vector.
- The **data types** of a vector determine what you can do with it
- Any time you are not sure what something does you can use `?` or `help()`

HOMEWORK

- Revise R_exercise_0.R, R_exercise_1.R + R_exercise_2.R as homework

R_exercise_2.R

- Use swirl to revise and practice what you learned
- Read sections 1.1 to 1.10 of “Statistics for Linguists: An Introduction Using R”
  - https://appliedstatisticsforlinguists.org/bwinter_stats_proofs.pdf
- Read this tutorial and try to load data into R
  - https://flowingdata.com/2015/02/18/loading-data-and-basic-formatting-in-r/